

5 Genetic screening can be used to test for aneuploidy.

Aneuploidy is the presence of an abnormal number of chromosomes in a cell.

Aneuploidy can affect the miscarriage rate of implanted embryos.

Following screening, only embryos with the correct number of chromosomes are implanted into the female.

The table shows the miscarriage rate of two groups of implanted embryos:

- embryos not screened for aneuploidy
- embryos screened and shown not to have aneuploidy.

Age range of women at implantation / years	Miscarriage rate (%)	
	Implanted embryos not screened for aneuploidy	Implanted embryos screened and shown not to have aneuploidy
<35	12.0	11.2
35 to 37	16.8	13.0
38 to 40	25.0	13.6
41 to 42	37.9	16.3
>42	58.8	17.2

- (a) Explain how this data shows that there is a correlation between the age of the women and the miscarriage rate.

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- (b) (i) Explain the conclusions that can be made from these data about the causes of miscarriage.

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- (ii) Explain why conclusions made using these data may not be valid.

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(c) Discuss the implications of screening embryos for aneuploidy before implantation.

(3)

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(Total for Question 5 = 10 marks)

